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Individual document: Variability area

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**Next-GenScientist: Blazar Variability and Time-Series
Analysis in Gamma-Ray Astronomy**

0.1 Introduction and own experience

Joining this project turned out to be a grateful experience where I had the opportunity to learn many new concepts in statistics and coding. It pushed me to connect the dots between the abstract world of theoretical astrophysics and the hands-on challenges of data science.

I spent most of my time exploring methods to analyze and identify flares, focusing on the seismic-inspired STA/LTA detection algorithm applied to real observational data. At the same time, I immersed myself in the scientific literature particularly Burd et al. (2021) to better understand the Ornstein Uhlenbeck (OU) stochastic process. Along the way, I wrestled with code debugging, local execution issues, and the mathematical rigor of mean-reverting of the OU Stochastic Differential Equation (SDE).

0.2 General Problem and Objectives

Previously, detection efforts relying on simple ratio thresholds within the Short-Term Average over Long-Term Average (STA/LTA) algorithm were abandoned because they lacked clear event definitions and statistical corrections, yielding erratic false positive rates. Then, the team has tried other methods such as the Ornstein-Uhlenbeck stochastic differential model.

Furthermore, the primary objectives undertaken to address this challenge were:

- **Literature and Theoretical Review:** Study foundational papers, specifically Burd et al. (2021) (“Ornstein-Uhlenbeck parameter extraction from light curves of Fermi-LAT observed blazars”), to understand stochastic representations of blazar light curves.
- **Assimilating the Ornstein-Uhlenbeck (OU) Framework:** Comprehend the analytical formulation of the continuous mean-reverting stochastic process driven by stochastic differential equations (SDE), incorporating mean flux μ , reversion rate θ , and fluctuation amplitude σ .
- **Bridging Stochastic Simulation and Algorithmic Detection:** Formulate a framework connecting the theoretical OU process (as a synthetic control benchmark) to time-series event detection (the STA/LTA algorithm) to calibrate false alarm rates and establish experimental ground truth.

0.3 Individual Role, Task Allocation, and Specific Focus

Talking about my individual role in the variability area, my efforts were focused on learning how these methods work. For example, I first explored the STA/LTA seismology method, where I also learned about Bayesian blocks. Later, I dedicated most of my time to understanding how the Ornstein-Uhlenbeck (OU) method works. First, I had to study an introduction to stochastic differential equations (SDEs), investigating on my own topics such as Brownian motion, the Wiener process, and a general idea of how SDEs work. In consequence, then I read the paper “Ornstein-Uhlenbeck parameter extraction from light curves of Fermi-LAT observed blazars” where I put all these concepts together to adapt them in a astrophysics context.

In addition, I found that the Ornstein-Uhlenbeck method is useful in areas such as: finance, biology, geophysics and in general physics/astrophysics.

More things about the OU paper

According about what I studied and read, a general idea about the paper is the following:

The Ornstein-Uhlenbeck model is used to describe processes that evolve randomly over time but tend to fluctuate around a long-term average value. This property is known as mean reversion.

What this method aims to demonstrate

According to the paper, the goal was to test whether the Ornstein-Uhlenbeck (OU) stochastic process is sufficient to perfectly reproduce all the statistical and physical properties observed in the real light curves of blazars. To do this, the method simulates those light curves to study their physical behavior and then compares them with real light curves; in this case, from 236 blazars observed by the FERMI telescope, which orbits 550 km above Earth. These blazars were monitored for a decade using monthly time bins. In the end, the comparison showed that the simulated light curves matched the real ones, and therefore it was argued that this is a valid method to consider.

0.4 Current State: Where the Work Was Left Off

In general, a solid theoretical foundation has been established for comparing stochastic continuous modeling (Log-OU) against fixed-window algorithms (STA/LTA) and adaptive partition methods (Bayesian Blocks + HOP) within the Variability area.

0.5 Areas for Improvement and Future Work

This document provides a general overview of the data treatment, mathematical modeling, and event curation performed to characterize blazar variability in Markarian 421 (Mrk 421) and Markarian 501 (Mrk 501) using observations from the FACT and HAWC observatories. An appendix has been included at the very end of this work to explain some relatively important functions and mathematical formulations used in the code shared via GitLab and Jupyter Notebooks. This document isn't focused on explaining each code line of the project's efforts. However, it is a great tool that the research team has created to help future young researchers catch up quickly and target specific questions.

0.6 Current State: Where the Work Was Left Off

Looking back, I feel the project reached a clear stopping point. I managed to get the flare detection function running smoothly in my local setup, producing a first draft of an event table for Mrk 421 that marks when flares start, peak, and end. Beyond that, I built a solid understanding of how different approaches compare—whether it's continuous stochastic modeling, fixed-window algorithms, or adaptive methods like Bayesian Blocks. Furthermore, I studied a simply code to simulate the Ornstein-Uhlenbeck method that may help future students researchers to understand its applications in flare detections.

0.7 Areas for Improvement and Future Work

There are still several directions that could make this work stronger if someone picks it up later. One would be to cross-check the flare detections with data from other instruments like HAWC, to make sure the signals aren't just artifacts from a single telescope. Another would be to build a complete simulation pipeline that runs synthetic data through the detection function, so false positives can be mapped out more systematically. Finally, it would be valuable to zoom in on shorter time scales like 20-minute bins during flare activity to better capture intraday variability and add that detail into the catalog.

1 Appendix: Key references and Mathematical Formulations used during this project:

References

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